Wildfire Evacuations and First Nations Peoples

1 Introduction

1.1 Wildfires, Evacuation and First Nations Peoples

There are roughly 7,000 wildfires per year in Canada, leading to over 1000 evacuations. While wildfires are a natural part of the ecosystem, human activity has exacerbated the problem through fires caused directly by humans and by climate change Natural Resources Canada (2025). They can have devastating impacts on people and the environment through damage to property, ecosystems and loss of life. The study of modelling, predicting, evaluating and mitigating the impacts of wildfires is thus an increasingly vital area of research.

While many wildfires are occur in unpopulated areas, sometimes they are sufficiently close to humans that evacuation is required. Wildfire evacuation is a complex and challenging process with many competing interests and unique aspects to every fire. For example, the BC Wildfire Service strives to balance saving lives and damage with risk and lives of responders Government of British Columbia (2026). However, evacuation decisions are inevitably subject to the flaws and bias of every institutional decision-making process.

Canada is home to over 3,000 first nation (FN) reserves Crown-Indigenous Relations and Northern Affairs Canada (2023). Such reserves are close to densely populated cities, but many are remote. Thus, small populations of FN peoples can be particularly vulnerable to risks posed by wildfires. Indeed, FN people consist of just 5% of the population but since 1980, 36% of evacuations have been from FN reserves. Further, Mihalus et al. (2023a) highlight challenges in decision-making, including limited Indigenous participation and a lack of local control over evacuation processes, which may contribute to inequitable outcomes. Therefore, studying the relationship between wildfires, evacuation and FN peoples is an important area of research to understand the risks and impacts of wildfires on FN peoples, and to inform policies and practices to mitigate these impacts.

This study aims to develop a Bayesian probabilistic model of the evacuation decision process by drawing on data from wildfires, evacuations and FN reserves in order to identify potential bias in the evacuation process. The remainder of the introduction discusses related work and describes the data used. Section 2 introduces various probabilistic models and compares their performance. Section 3 interprets the results and analyses their implications for wildfire evacuation and FN peoples. Finally, Section 4 concludes and discusses future work.

1.2 Related Work

Christianson et al. (2024) provide a comprehensive, up-to-date descriptive analysis of wildfire evacuations in Canada, including trends over time, geographic distribution and causes. Their conclusions are broad. With regards to FN peoples, they find that they are disproportionately affected by wildfire evacuations, with 14 of the 16 communities in Canada that experienced the most wildland fire evacuations were FN. McGee et al. (2022) analyse over 1000 evacuations by province, local region, time of year and cause to identify three 'syndromes' that cover almost 80% of all evacuations: (1) lightning-caused fires in summer in remote areas, (2) less remote human-caused fires in summer, and (3) interior BC fires in summer. They also analyse evacuations in relation to FN reserves and find that effects and abundance differ across regions. Their work is a useful descriptive analysis of the data, but does not attempt to model the evacuation process. Mihalus et al. (2023b) provide a more qualitative analysis of the evacuation process. Their contributions regarding FN peoples focuses on the bias and challenges communities face after evacuation.

Thus, a predictive model of evacuation decisions is a useful contribution to the literature. It can be used to identify correlations between wildfire and evacuation characteristics, and to identify potential bias in the evacuation process.

1.3 Data & Code

Data: Data on wildfire from the Canadian National Fire Database Canadian Forest Service (2024) was joined with data on wildfire evacuations from the Canadian Wildfire Evacuation Database Beverly and Bothwell (2011). This join was done based on proximity in space and time. The fire data contains information on the location, size, cause and other characteristics of wildfires. The evacuation data contains information on the location, date and other characteristics of evacuations. Geospatial data on FN reserves from the Canadian Geospatial Data Infrastructure Crown-Indigenous Relations and Northern Affairs Canada (2023) was used to compute distance to nearest FN reserve and whether the fire was in a FN reserve. Categorical variables were simplified and much of the data was dropped. Further details are given in Appendix C and in the code.

The final dataset contains $N = 15,203$ fires, of which 531 had at least one associated evacuation and 269 were on FN reserves. Predictors are: `province` (province of fire, $k = 12$), `response_type` (management response level, $k = 4$), `protection_zone` (land management category, $k = 8$), `fn_indicator` (fire in FN reserve), `log_dist_to_fn_km` (standard scaled log distance to nearest reserve), `log_fire_size_ha` (standard scaled log fire size in hectares) and `region` (categorical variable indicating the grid cell over Canada, defined as squares of approximately $3^\circ \times 3^\circ$ in latitude and longitude, $k = 145$ regions with wildfire data). The binary target is `any_evacuation` (whether any evacuation occurred).

Code: All models were created in STAN and compiled and analysed in python using cmdstanpy. I used 4 chains, 2000 iterations and a warmup of 1000. Analysis of mixing, stability and performance was done using Arviz and can be found in Section 2. All code and data can be found on GitHub and an example of Stan code used can be found in Appendix F.

2 Probabilistic Modelling and Inference

2.1 Logistic Regression

I use a series of increasingly complex logistic regression models, each of the following form:

$$\theta_i = \alpha + \alpha_{\text{cat}}(\mathbf{c}_i) + \boldsymbol{\beta}^\top \mathbf{x}_i, \qquad y_i \sim \text{Bernoulli}(\text{logistic}(\theta_i))$$

with priors

$$\alpha_{\text{cat}}(j) \sim \mathcal{N}(0,1) \quad \forall\, j \in \text{cat}, \qquad \boldsymbol{\beta} \sim \mathcal{N}(0,1), \qquad c \sim \mathcal{N}(0,2)$$

where $\mathbf{x}_i$ denotes continuous covariates and $\alpha_{\text{cat}}(\mathbf{c}_i)$ denotes the contribution of categorical variables (e.g. province or protection zone), j is a value for a given category cat, and c is the intercept. Priors were chosen to have full support with slightly more variation for c to allow for a low baseline evacuation probability. A hierarchical model could have been employed with, but I found this increased computational complexity while $N = 15,203$ was sufficient to estimate separate category effects directly. Below, I state the changes of each subsequent model to the previous as they are a series of nested models.

Minimal: only fire size and whether the fire is on a FN reserve.

$$\mathbf{x}_i = \big(\log \text{fire_size},\ \text{fn_indicator}_i\big), \qquad \alpha_{\text{cat}}(\mathbf{c}_i) = 0$$

Provinces: add provinces to the categorical variables.

$$\alpha_{\text{cat}}(\mathbf{c}_i) = \alpha_{\text{prov}[i]}$$

Full: add distance to nearest FN reserve and the protection zone variables.

$$\mathbf{x}_i = \big(\log \text{fire_size},\ \text{log_dist_to_fn}_i,\ \text{fn_indicator}_i\big), \qquad \alpha_{\text{cat}}(\mathbf{c}_i) = \alpha_{\text{prov}[i]} + \alpha_{\text{prot}[i]}$$

Responses: add the response type to the categorical variables.

$$\alpha_{\text{cat}}(\mathbf{c}_i) = \alpha_{\text{prov}[i]} + \alpha_{\text{prot}[i]} + \alpha_{\text{resp}[i]}$$

Spatial (VI): remove response type and add region. This model was too computationally expensive to run inference using MCMC thus I used variational inference (VI) to approximate the posterior using the default cmdstanpy parameters ('meanfield' algorithm, independence assumption).

$$\mathbf{x}_i = \big(\log(\text{fire_size}_i),\ \text{log_dist_to_fn}_i,\ \text{fn_indicator}_i\big), \qquad \alpha_{\text{cat}}(\mathbf{c}_i) = \alpha_{\text{prov}[i]} + \alpha_{\text{prot}[i]} + u_{\text{region}[i]}.$$

with a spatial prior imposed on u_{region}. Let E denotes the set of neighbouring regions. The prior imposes the belief that neighbouring regions have related rates of evacuations.

$$u_{\text{region}} \sim \mathcal{N}(0,1), \qquad \sum_{r=1}^{R} u_r = 0, \qquad p(\mathbf{u}) \propto \exp\left(-\frac{1}{2\sigma^2_{\text{region}}} \sum_{(r,s)\in E} (u_r - u_s)^2\right),$$

Diagnostics: I inspected trace and rank plots for the MCMC models. An example (full model, parameters α and β_{fn}) is shown in Appendix D, Figure 1. I also ensured that $\hat{R} \approx 1$ for all parameter values indicating that all chains agree, and found that the effective sample size (ESS) was over 700 for all parameters on all models in both tail and bulk, with most values between 1000 and 3000, indicating good convergence. I ran posterior predictive checks (PPC) for each model, verifying that a credible interval over the total number of evacuations is consistent with the data. These can be seen below in Table 1. Finally, I ran a brief sensitivity analysis on the prior parameters for the Province model in Appendix E which indicated that parameter means were stable across priors.

2.2 Model Comparison

I compare the nested logistic models using Leave-one-out (LOO) expected log predictive density (ELPD). That is, ELPD $= \sum_{i=1}^{N} \log p(y_i|y_{-i})$. Higher ELPD indicates better out-of-sample predictive performance. Table 1 reports ELPD with its standard error (SE), the ELPD difference from the best model, the standard error of that difference (DSE) and a 95% credible interval (CI) for total number of evacs. All Pareto-k values used for estimation of the ELPD were in the 'good' category (< 0.7) indicating stability of the estimation.

Table 1: Model comparison using LOO ELPD.

Model	ELPD ± SE	ELPD diff	DSE	Mean total evacs [95% CI]
Response	**-1865** ± 63.0	0.0	0.0	532 [475, 593]
Spatial (VI)	-1868 ± 63.3	2.8	9.2	538 [448, 636]
Full	-1871 ± 63.1	5.6	3.5	532 [473, 592]
Province	-2019 ± 66.5	154.0	17.6	532 [471, 595]
Minimal	-2247 ± 73.8	381.8	28.8	532 [472, 595]

All models posterior CI's for the total number of evacuations are sensible, with the true value at 531. Note the Spatial model with VI has a wider CI interval and mean slightly farther than the rest from the true value. Response gives the best ELPD, while Spatial (VI) and Full are close with differences only 1 or 2 SEs apart. There is a large and meaningful gap between Province and the top three models, and an even larger one between Minimal and the rest.

3 Results & Analysis

Table 2 reports posterior means and 95% credible intervals for key parameters across all fitted models. Entries marked — indicate that the parameter is not present in that model specification.

Table 2: Posterior mean and 95% CI for selected parameters across models.

Model	α	β_{fn}	$\beta_{\text{fire_size}}$	$\alpha_{\text{prov=BC}}$	$\beta_{\text{dist_to_fn}}$
Minimal	$-3.4_{[-3.5,\,-3.3]}$	$1.4_{[1.0,\,1.8]}$	$0.4_{[0.3,\,0.5]}$	—	—
Province	$-4.0_{[-4.7,\,-3.4]}$	$1.2_{[0.8,\,1.6]}$	$0.5_{[0.4,\,0.6]}$	$1.8_{[1.2,\,2.5]}$	—
Full	$-3.9_{[-4.9,\,-2.9]}$	$-1.0_{[-1.4,\,-0.5]}$	$0.5_{[0.4,\,0.6]}$	$1.5_{[0.9,\,2.2]}$	$-1.1_{[-1.2,\,-1.0]}$
Response	$-4.0_{[-5.3,\,-2.6]}$	$-1.0_{[-1.4,\,-0.5]}$	$0.5_{[0.4,\,0.6]}$	$1.5_{[0.9,\,2.1]}$	$-1.1_{[-1.2,\,-0.9]}$
Spatial (VI)	$-2.6_{[-2.7,\,-2.5]}$	$-0.9_{[-1.3,\,-0.5]}$	$0.5_{[0.5,\,0.6]}$	$0.4_{[0.2,\,0.6]}$	$-1.0_{[-1.1,\,-0.9]}$

Intercept: All model broadly agree on the value for α, with Minimal a small outlier and Spatial a larger outlier. Notably the 95% CI for the Spatial model is much tighter than the rest. As a quick

validation, I ran the Full model with VI and found a posterior for α of -2.28 with CI $[-2.4, -2.2]$. However, as with the Spatial model, other parameter values were more consistent with the MCMC inference. It may be that the VI approximation for α is a particularly bad estimate of the posterior. From this, I conclude that we can give some weight to the Spatial model, especially if they agree with the other models, but not too much as the VI approximation certainly loses information.

FN-Indicator & Distance to FN Reserve: The most notable result here is the stark change from a positive to negative β_{fn} between the Minimal/Province models and the remaining models. This is likely due to the presence of the $\beta_{\text{dist_to_fn}}$ in the final 3 models. However, its value is negative as is the case for β_{fn} which is unsurprising. It is reasonable that the farther the fire is from a reserve, the less likely an evacuation is to be ordered. However, β_{fn} remains negative, suggesting that a fire being on the reserve is roughly associated with a $\exp(-1) = 0.37$ reduction in odds, or 67% less of an evacuation. Note that these results are purely associative and causal.

Provinces: The province with the highest α was always British Columbia (BC) with mean $\alpha_{\text{BC}} \approx 1.5$ for the full model, while Quebec (QC) had $\alpha_{\text{QC}} \approx -1.4$. In general, these coefficients suspiciously correlate with the provinces which have the most fires. It may be that provinces with the most wildfires are more experienced in handling evacuations, are more risk-averse as there is greater potential for the wildfire to grow, or that the model learns that some provinces almost never have enough fires large enough to issue an evacuation. Regardless of the cause, this suggests there are confounders missing about the evacuation practices of each province.

Responses: This model was intended to test the impact of including response severity in the model on β_{fn}. In a sense, this model is unfair since perfect knowledge of the response could include the evacuation decision. However, the model obtains very similar ELPD and parameter means as the Full model. Indeed, $\beta_{\text{fn}} \approx -1$ for both models. However, the only response α with a positive coefficient is the 'other', corresponding to Nans in the original data. Meanwhile a 'full' response has a mean of -0.05 with a standard deviation (SD) of 0.52 indicating relatively high uncertainty.

Protection Zones: The protection zones show a clearer effect. In the Full model, the 'nordique' zone has the largest mean of 1.42 and SD of 0.61, while the smallest is 'monitor' with a mean of -1.52.

Regions & Spatial Model: The regional effects vary from -1.43 to 1.1 where the mean of σ_{region} is 3.65. This model should be able to isolate further confounders like 'remoteness', while retaining regularisation using a prior. However, the model obtains a very similar ELPD as the Full model. This may be in part due to the VI method. The value for α_{BC} decreases meaningfully to a mean of 0.4. I assume that much of the 'BC effect' has instead been captured by the smaller regional factors in the model. Crucially, the value of β_{fn} remains negative in this model.

4 Conclusion

This study has looked at a series of probabilistic models to investigate correlations between factors of wildfires and their surrounding communities and the order of an evacuation. The change from positive to negative coefficient on the FN indicator variable after introducing further covariates suggests that FN reserves are correlated with a lower likelihood of an evacuation order. The stability of this conclusion after giving the model access to response severity and controlling for 'remoteness' using a grid of regions adds weight to the claim. This could suggest bias in the decision making process fuelled by priorities embedded in settler systems of power, but the results observed here should not be taken in isolation as suggesting a causal relationship. Other statistical methods and qualitative analysis of such systems should be used to investigate causality.

Limitations: Among the main limitations is the data. The join between fires and evacuations is imperfect, and the removal of unmatched evacuations may bias results. Further, the data is observational and thus subject to confounding and bias. Ideas around 'remoteness' are hard to capture, even by local regions, and may be a meaningful confounder. The model is also limited in its ability to capture the complex decision-making process around evacuation, which may involve many unobserved factors. Further, not all fires are recorded in the data, and not all evacuations are recorded, which may bias results. A 'hurdle' model that accounts for the fact that some fires are not reported may be a useful extension.

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Appendix

A Positionality and Land Acknowledgement Statement

I am a white male settler...

B AI Usage Statement

ChatGPT was used at various points in this study for finding relevant sources, debugging code, suggesting modelling ideas and code for generating plots. It was not used for writing, analysis or used to directly generate any results. All code was also initially written and run by me directly, with AI assistance only used for debugging or understanding library API's.

C Data

C.1 Original Data

Canadian Forest Service (2024) **Fire Data:**

- `NFDBFIREID`, `FIRE_ID`, `FIRENAME`: unique identifiers and name
- `SRC_AGENCY`, `NAT_PARK`: reporting agency and park indicator
- `LATITUDE`, `LONGITUDE`: fire location
- `YEAR`, `MONTH`, `DAY`: ignition date components
- `REP_DATE`, `OUT_DATE`: reported and extinguished dates

- SIZE_HA: fire size (hectares)
- CAUSE: ignition cause (e.g. human, natural)
- FIRE_TYPE: fire classification
- RESPONSE: management response level
- PROTZONE: protection zone classification
- MORE_INFO: additional notes
- FID, the_geom: internal ID and geometry

Beverly and Bothwell (2011) **Evacuation Data:**

- EvacID: unique evacuation identifier
- Year, Province, Region: temporal and geographic context
- Location, Lat, Long: evacuation location
- Population, PopulationNotes: affected population
- FN_Reserve, FN_Reserve_num: First Nations reserve indicators
- EvacSize: evacuation scale
- IssueDate, EndDate: evacuation duration
- EvacReason: stated cause of evacuation
- FireSource, FireIgnit, FireSize, RepDOY: linked fire attributes
- Group: grouping/category field
- matched: indicator for linkage to fire dataset

Crown-Indigenous Relations and Northern Affairs Canada (2023) **First Nations Reserve Data:**

- geometry: spatial polygon defining reserve boundaries
- NAME1–NAME5, LANGUAGE1–LANGUAGE5: reserve names in multiple languages
- ALTYPE: reserve type (e.g. Indian Reserve, Land Claim)
- JUR1–JUR4: jurisdictional information
- ALCODE, NID: identifiers
- PROVIDER, DATASETNAM, SPECVERS: dataset metadata
- CREDATE, REVDATE: creation and revision dates
- ACCURACY, ACQTECH, METACOVER: data quality and acquisition details
- WEBREF: external reference link

C.2 Data Cleaning & Joining

Fire and evacuation records were linked using a nearest-neighbour heuristic in both time and space. For each evacuation, candidate fires were restricted to those reported within a ± 14 day window of the evacuation issue date, and the closest such fire (in geodesic distance) was selected, provided it lay within 100 km. Evacuations without a valid match under these criteria were discarded, resulting in approximately three quarters of evacuations being retained. However, this introduces potential bias: unmatched evacuations were disproportionately associated with First Nations reserves, so their removal may attenuate estimated effects related to Indigenous proximity. This is a limitation of this matching method that could be investigated by further work.

Geospatial joins were then used to augment fire records with First Nations reserve features. Fires were tagged with a binary fn_indicator based on whether their location intersected a reserve polygon, and continuous measures such as log_dist_to_fn_km (distance to the nearest reserve) and n_fn_20km (count of reserves within a 20 km radius) were computed using projected geometries.

The resulting fire-level dataset was aggregated so that multiple evacuations linked to the same fire contributed summary statistics (e.g. counts, total population, and mean match distance), and missing values for unmatched fires were filled with zeros or False as appropriate.

Finally, categorical variables were simplified and standardised to provide a smaller number of categories, continuous variables (fire size and distance to nearest FN reserve) were log-transformed and normalised, and irrelevant fields were removed. Three rows causing numerical instability in posterior inference (due to near-deterministic predictions) were excluded as a pragmatic step.

D Trace & Rank Plots

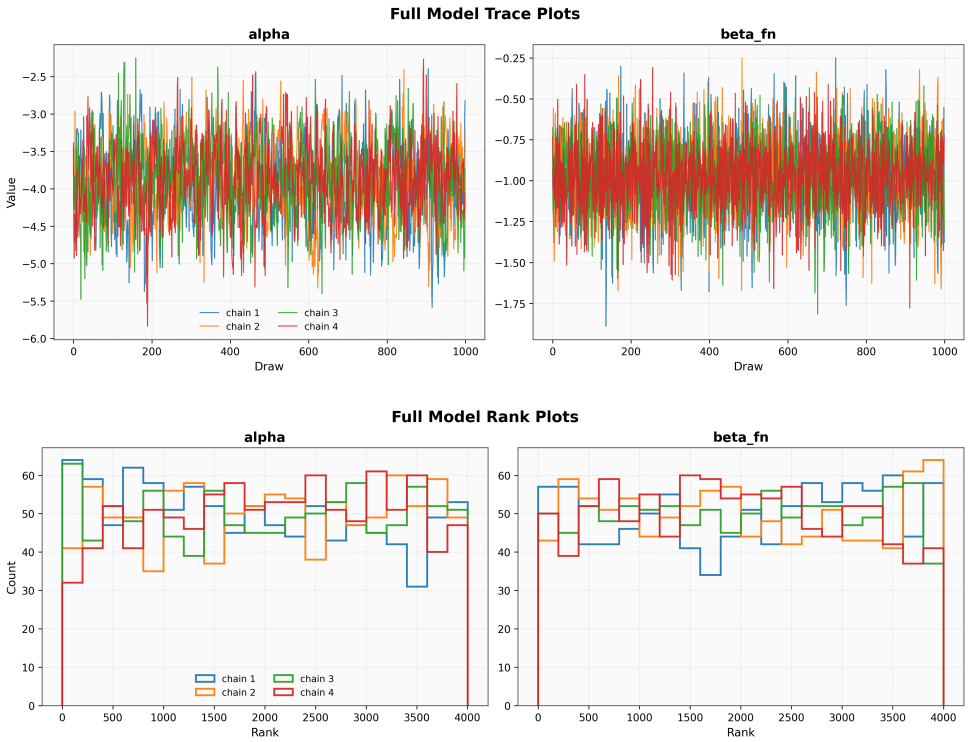

Figure 1: Full-model MCMC diagnostics for parameters α and β_{fn}: trace plots (top) and rank plots (bottom). Plots indicate fast mixing and stable estimations of parameter values.

E Sensitivity Analysis

Sensitivity analysis was done only on the Province model with 3 different prior parameter settings for reasons of computational cost. The results are summarised in Table XX below.

F Example Stan Code

Below is the Stan code used for the Minimal model. Note that full code for all models, compilation and analysis can be found on Github[1].

```
data {
  int<lower=1> N;
  array[N] int<lower=0, upper=1> y;
  vector[N] log_fire_size;
  array[N] int<lower=0, upper=1> fn_indicator;
}

parameters {
```

[1]https://github.com/PierreRL/Wildfire_Evacuations

```
  real alpha;
  real beta_log_fire_size;
  real beta_fn;
}

model {
  vector[N] theta;

  // Priors
  alpha ~ normal(0, 2);
  beta_log_fire_size ~ normal(0, 1);
  beta_fn ~ normal(0, 1);

  for (n in 1:N) {
    theta[n] = alpha
             + beta_log_fire_size * log_fire_size[n]
             + beta_fn * fn_indicator[n];
  }

  // Likelihood
  y ~ bernoulli_logit(theta);
}

generated quantities {
  vector[N] log_lik;
  array[N] int y_rep;

  for (n in 1:N) {
    real theta_n = alpha
                 + beta_log_fire_size * log_fire_size[n]
                 + beta_fn * fn_indicator[n];
    log_lik[n] = bernoulli_logit_lpmf(y[n] | theta_n);
    y_rep[n] = bernoulli_logit_rng(theta_n);
  }
}
```